and

$$\mathbf{X}^0 \stackrel{\text{def}}{=} \left(\mathbf{x}_0^0; \mathbf{x}_1^0; \cdots; \mathbf{x}_T^0\right). \tag{91}$$

These will be the inputs to the transformer layers and thus,[33]

$$\mathcal{R}\left(\boldsymbol{y}_{<t}\right) \stackrel{\text{def}}{=} \left(\mathbf{x}_0^0; \mathbf{x}_1^0; \cdots; \mathbf{x}_{t-1}^0\right). \tag{92}$$

With this, we can describe and prove the correctness of the first layer of the transformer we are building.

Lemma F.6 (Layer 1)**.** *There exists a transformer layer $\mathcal{L}_1$ that, given the inputs $\mathbf{X}^0$, computes the values $\frac{c(t)}{t+1}$ and $\frac{c(t-1)}{t+1}$ for all $t = 1, \ldots, T$. More precisely, denoting*

$$\mathbf{X}^1 \stackrel{\text{def}}{=} \left(\mathbf{x}_0^1; \mathbf{x}_1^1; \cdots; \mathbf{x}_T^1\right) = \mathcal{L}_1\left(\mathbf{X}^0\right), \tag{93}$$

it holds that $\mathbf{x}_t^1$ contains the entries containing the values $\frac{c(t)}{t+1}$ and $\frac{c(t-1)}{t+1}$:

$$\mathbf{x}_t^1 = \begin{pmatrix} [\![q_t]\!] \\ [\![v_{t-1}]\!] \\ [\![a_{t-1}]\!] \\ a_{t-1}, a_{t-2} \\ \frac{c(t)}{t+1}, \frac{c(t-1)}{t+1} \\ 0, \mathbf{0}_{|\Gamma|} \\ 1, t+1, \frac{1}{t+1}, \frac{1}{(t+1)^2} \end{pmatrix} \tag{94}$$

Proof. This follows from Pérez et al. (2021, Lemma 9), which we summarize here.[34] Concretely, $\mathcal{L}_1$ is implemented by a transformer layer with trivial (zero-valued) query and key transformations and a value transformation V that copies the values of the actions a_{t-1} and a_{t-2} to the entry that will hold the values $\frac{c(t)}{t+1}$ and $\frac{c(t-1)}{t+1}$. Since the current head location $c(t)$ is simply the sum of those values (cf. Eq. (86)), attending to all previous positions results in Eq. (94).

Formally, we define

$$\mathbf{Q}^1 \stackrel{\text{def}}{=} \mathbf{0}_{D\times D}, \quad \mathbf{K}^1 \stackrel{\text{def}}{=} \mathbf{0}_{D\times D}, \tag{95a}$$

$$f(\mathbf{q}, \mathbf{k}) \stackrel{\text{def}}{=} \langle \mathbf{q}, \mathbf{k} \rangle \tag{95b}$$

$$\mathbf{V}_{n',:}^1 = \mathbf{V}_{n,:}^1 = \begin{pmatrix} \mathbf{0}_{|Q|} \\ \mathbf{0}_{|\Gamma|} \\ \mathbf{0}_{|A|} \\ 0, 0 \\ 1, 1 \\ 0, \mathbf{0}_{|\Gamma|} \\ 0, 0, 0, 0 \end{pmatrix}^\top \tag{95c}$$

$$\mathbf{O}^1 = \mathbf{0}_{D\times D} \tag{95d}$$

Here, n' and n refer to the indices of the rows at which the values $\frac{c(t)}{t+1}$ and $\frac{c(t-1)}{t+1}$ will be stored (that is, the rows below the values of a_{t-1} and a_{t-2}). All other rows of $\mathbf{V}^1$ are zero.

Then, for $t \in \mathbb{N}_{\geq 0}$, it holds that

$$f(\mathbf{q}_t, \mathbf{k}_j) = f\left(Q\left(\mathbf{x}_t^0\right), K\left(\mathbf{x}_j^0\right)\right) = f\left(\mathbf{Q}^1\mathbf{x}_t^0, \mathbf{K}^1\mathbf{x}_j^0\right) = \langle \mathbf{0}, \mathbf{0} \rangle = 0 \tag{96}$$

[33] Note that since the transformer over the augmented alphabet is deterministic, this representation is unique.

[34] More precisely, since their construction stores the actions a_t and a_{t-1} (the former is possible because the action is not sampled but deterministically computed based on the configuration of the PTM), their layer computes the values $\frac{c(t)}{t+1}$ and $\frac{c(t)}{t+1}$. As we show later in Lemma F.7, this does not affect the correctness of the construction.

for all $j \leq t$, resulting in $\mathbf{s}_t = \mathrm{hardmax}\left(\mathbf{0}\right) = \frac{1}{t+1}\mathbf{1}_t$, and

$$V\left(\mathbf{x}_j^0\right) = \begin{pmatrix} \mathbf{0}_{|Q|} \\ \mathbf{0}_{|\Gamma|} \\ \mathbf{0}_{|A|} \\ 0, 0 \\ a_{j-1}, a_{j-2} \\ 0, \mathbf{0}_{|\Gamma|} \\ 0, 0, 0, 0 \end{pmatrix}. \tag{97}$$

This results in

$$\begin{aligned}
\texttt{Att}\left(\mathbf{q}_t, \mathbf{K}_t, \mathbf{V}_t\right) &= \sum_{j=0}^{t} s_j V\left(\mathbf{x}_j^0\right) && \text{(98a)} \\
&= \sum_{j=0}^{t} \frac{1}{t+1} V\left(\mathbf{x}_j^0\right) && \text{(98b)} \\
&= \frac{1}{t+1} \sum_{j=0}^{t} V\left(\mathbf{x}_j^0\right) && \text{(98c)} \\
&= \frac{1}{t+1} \sum_{j=0}^{t} \begin{pmatrix} \mathbf{0}_{|Q|} \\ \mathbf{0}_{|\Gamma|} \\ \mathbf{0}_{|A|} \\ 0, 0 \\ a_{j-1}, a_{j-2} \\ 0, \mathbf{0}_{|\Gamma|} \\ 0, 0, 0, 0 \end{pmatrix} && \text{(98d)} \\
&= \frac{1}{t+1} \begin{pmatrix} \mathbf{0}_{|Q|} \\ \mathbf{0}_{|\Gamma|} \\ \mathbf{0}_{|A|} \\ 0, 0 \\ c\left(t\right), c\left(t-1\right) \\ 0, \mathbf{0}_{|\Gamma|} \\ 0, 0, 0, 0 \end{pmatrix} && \text{(98e)} \\
&= \begin{pmatrix} \mathbf{0}_{|Q|} \\ \mathbf{0}_{|\Gamma|} \\ \mathbf{0}_{|A|} \\ 0, 0 \\ \frac{c(t)}{t+1}, \frac{c(t-1)}{t+1} \\ 0, \mathbf{0}_{|\Gamma|} \\ 0, 0, 0, 0 \end{pmatrix}. && \text{(98f)}
\end{aligned}$$

Furthermore, we have that

$$\mathbf{a}_t = \texttt{Att}\left(\mathbf{q}_t, \mathbf{K}_t, \mathbf{V}_t\right) + \mathbf{x}_t^0 = \begin{pmatrix} \mathbf{0}_{|Q|} \\ \mathbf{0}_{|\Gamma|} \\ \mathbf{0}_{|A|} \\ 0, 0 \\ \frac{c(t)}{t+1}, \frac{c(t-1)}{t+1} \\ 0, \mathbf{0}_{|\Gamma|} \\ 0, 0, 0, 0 \end{pmatrix} + \begin{pmatrix} [\![q_t]\!] \\ [\![v_{t-1}]\!] \\ [\![a_{t-1}]\!] \\ a_{t-1}, a_{t-2} \\ 0, 0 \\ 0, \mathbf{0}_{|\Gamma|} \\ 1, t+1, \frac{1}{t+1}, \frac{1}{(t+1)^2} \end{pmatrix} = \begin{pmatrix} [\![q_t]\!] \\ [\![v_{t-1}]\!] \\ [\![a_{t-1}]\!] \\ a_{t-1}, a_{t-2} \\ \frac{c(t)}{t+1}, \frac{c(t-1)}{t+1} \\ 0, \mathbf{0}_{|\Gamma|} \\ 1, t+1, \frac{1}{t+1}, \frac{1}{(t+1)^2} \end{pmatrix} \tag{99}$$

$$\mathbf{x}_t^1 = O\left(\mathbf{a}_t\right) + \mathbf{a}_t = \mathbf{0}_D + \begin{pmatrix} [\![q_t]\!] \\ [\![v_{t-1}]\!] \\ [\![a_{t-1}]\!] \\ a_{t-1}, a_{t-2} \\ \frac{c(t)}{t+1}, \frac{c(t-1)}{t+1} \\ 0, \mathbf{0}_{|\Gamma|} \\ 1, t+1, \frac{1}{t+1}, \frac{1}{(t+1)^2} \end{pmatrix} = \begin{pmatrix} [\![q_t]\!] \\ [\![v_{t-1}]\!] \\ [\![a_{t-1}]\!] \\ a_{t-1}, a_{t-2} \\ \frac{c(t)}{t+1}, \frac{c(t-1)}{t+1} \\ 0, \mathbf{0}_{|\Gamma|} \\ 1, t+1, \frac{1}{t+1}, \frac{1}{(t+1)^2} \end{pmatrix}, \tag{100}$$

which is what we needed to show. ■

Lemma F.7 (Layer 2)**.** *There exists a transformer layer $\mathcal{L}_2$ that, given the outputs $\mathbf{X}^1$ of $\mathcal{L}_1$ from Lemma F.6, computes the values $\ell(t)$ and $[\![v_{\ell(t)}]\!]$ for all $t = 1, \ldots, T$. More precisely, denoting*

$$\mathbf{X}^2 \stackrel{\text{def}}{=} \left(\mathbf{x}_0^2; \mathbf{x}_1^2; \cdots; \mathbf{x}_T^2\right) = \mathcal{L}_2\left(\mathbf{X}^1\right), \tag{101}$$

it holds that $\mathbf{x}_t^2$ contains the entries containing the values $\ell(t) + 1$ and $[\![v_{\ell(t)}]\!]$:

$$\mathbf{x}_t^2 = \begin{pmatrix} [\![q_t]\!] \\ [\![v_{t-1}]\!] \\ [\![a_{t-1}]\!] \\ a_{t-1}, a_{t-2} \\ \frac{c(t)}{t+1}, \frac{c(t-1)}{t+1} \\ \ell(t) + 1, [\![v_{\ell(t)}]\!] \\ 1, t+1, \frac{1}{t+1}, \frac{1}{(t+1)^2} \end{pmatrix}. \tag{102}$$

Proof. This follows from Pérez et al. (2021, Lemma 10). The idea of the construction is for the self-attention mechanism at time t to attend to exactly the entry from time step $\ell(t) + 1$ (i.e., to compute the query and key vectors such that $\operatorname{argmax}(\mathbf{s}_t) = \ell(t) + 1$) since that entry will contain the information about the symbol written at the time step before—at $\ell(t)$. Then, the values $\ell(t)$ and $v_{\ell(t)}$ are obtained by copying the corresponding values of the positional encoding and the written symbol from that time step.